

Software Design Description & Code Review Specification

*Engineering Architecture for Slicing, Indexing, and Cloud Streaming the 1769
King James Bible*

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Executive Summary & System Architecture

Project Scope and Objective

The primary objective of the **Word of God** project is to transform an monolithic 1,437-page compilation of the King James Bible (1769 Authorized Version) into a modern, modular, and portable multi-volume digital library.

The architecture addresses three fundamental software challenges:

1. **Lossless Document Decomposition:** Extracting 66 canonical books into standalone PDF files with zero text overlap and zero distortion of typography.
2. **Dual-Tier Indexing (Local vs. Cloud):** Enabling instant navigation across volumes through both local inter-document filesystem links (/GoToR) and high-speed cloud-streamed endpoints (/URI).
3. **Cross-Platform Mobile Portability:** Overcoming mobile operating system (iOS/iPadOS) sandbox restrictions and MIME type download traps to deliver on-the-fly PDF streaming.

Architectural Component Matrix

The system consists of six distinct, production-verified solutions working in synergy:

Component	Source / Config	Output fact	Arti-	Core Role
A. PDF Slicer	split_bible.py	./books/*.pdf (67)		Automated bookmark extraction and boundary page slicing.
B. Local Master Index	Master_Index.tex	Master_Index.pdf		Offline desktop navigation using relative PDF action links.
C. Cloud Master Index	Master_Index_Cloud.tex	Master_Index_Cloud.pdf		Mobile/iCloud streaming via high-speed GitHub CDN links.
D. Web Bookshelf	index.html	index.html		Zero-dependency responsive web portal with real-time search.
E. Build Orchestrator	generate_indexes.py	Rebuilds B, C, D		Dynamic metadata aggregation and automated compilation.
F. PWA Audio Reader	manifest.json, sw.js	PWA App	Mobile	Fullscreen standalone mobile PWA with text reflow, synchronized verse audio, lock-screen controls, and offline caching.

Table 1: System Component Architecture Overview

Code Review: Solution A — Automated Boundary Extraction

File: split_bible.py

Purpose: Decompose KJV-Holy-Bible-1769.pdf (1,437 pages) into 66 individual book PDFs plus one standalone front-matter PDF.

```

1 import os, re
2 from pypdf import PdfReader, PdfWriter
3
4 PDF_PATH = "KJV-Holy-Bible-1769.pdf"
5 OUTPUT_DIR = "books"
6
7 def split_bible():
8     os.makedirs(OUTPUT_DIR, exist_ok=True)
9     reader = PdfReader(PDF_PATH)
10    outline = reader.outline
11    total_pages = len(reader.pages)
12
13    # 1. Front Matter Extraction (Pages 0 to 3)
14    first_book_page = reader.get_destination_page_number(outline[0])
15    if first_book_page > 0:
16        front_writer = PdfWriter()
17        for p in range(first_book_page):
18            front_writer.add_page(reader.pages[p])
19        with open(os.path.join(OUTPUT_DIR, "00_Front_Matter.pdf"), "wb") as f:
20            front_writer.write(f)
21
22    # 2. Sequential Book Range Extraction (Books 1 to 66)
23    for i, item in enumerate(outline):
24        book_num = i + 1
25        safe_title = re.sub(r"^[A-Za-z0-9_ -]", "", item.title).strip().replace(" ", "_")
26        filename = f"{book_num:02d}_{safe_title}.pdf"
27
28        start_page = reader.get_destination_page_number(item)
29        end_page = reader.get_destination_page_number(outline[i + 1]) - 1 if (i + 1 <
30        len(outline)) else (total_pages - 1)
31
32        writer = PdfWriter()
33        for p in range(start_page, end_page + 1):
34            writer.add_page(reader.pages[p])
35
36        with open(os.path.join(OUTPUT_DIR, filename), "wb") as f:
37            writer.write(f)

```

Design Analysis & Engineering Decisions

1. **Outline Tree Traversal vs. OCR/Regex:** Instead of unreliable text-scraping across 1,437 pages, the algorithm reads the PDF document's native Outline Dictionary (`/Outlines`). Inspection revealed that the original LaTeX compilation cleanly contained exactly 66 top-level book-marks.
2. **Zero-Indexed Boundary Calculation:** In PDF specifications, page destinations are 0-indexed. The start page of book $i + 1$ directly dictates the end page of book i via:

$$\text{end_page}_i = \text{start_page}_{i+1} - 1$$

For the final volume (Revelation), the upper boundary clamps to $\text{total_pages} - 1$.

3. **Canonical Sorting Preservation:** Book filenames are prefixed with two-digit zero-padded numbers (01_Genesis.pdf to 66_Revelation.pdf). This guarantees that operating systems and file managers list the books in canonical biblical sequence rather than alphabetical order.

Code Review: Solution B — Local Offline Master Directory

File: Master_Index.tex

Purpose: Provide a formal, 5-page master PDF that displays the original front matter and a two-column interactive directory linking to local files.

```

1 % Layout Architecture: Master_Index.tex
2 \documentclass[10pt,letterpaper]{article}
3 \usepackage[top=0.6in,bottom=0.6in,left=0.75in,right=0.75in]{geometry}
4 \usepackage{pdfpages}
5 \usepackage{tabularx}
6 \usepackage[hidelinks]{hyperref}
7
8 \begin{document}
9 % 1. Embed Original 4-page Front Matter
10 \includepdf[pages=-]{books/00_Front_Matter.pdf}
11
12 % 2. Page 5: Two-Column Single-Page Balanced Layout
13 \noindent
14 \begin{minipage}[t]{0.485\textwidth}
15   \centering{\bfseries\scshape The Old Testament \footnotesize(39 Books)}\\
16   \begin{tabularx}{\linewidth}{r X r}
17     01 & \href{books/01_Genesis.pdf}{Genesis} & 68 pp. \\
18     % ... books 2 through 39 ...
19   \end{tabularx}
20 \end{minipage}\hfill
21 \begin{minipage}[t]{0.485\textwidth}
22   \centering{\bfseries\scshape The New Testament \footnotesize(27 Books)}\\
23   \begin{tabularx}{\linewidth}{r X r}
24     40 & \href{books/40_Matthew.pdf}{Matthew} & 41 pp. \\
25     % ... books 41 through 66 ...
26   \end{tabularx}
27 \end{minipage}
28 \end{document}

```

Design Analysis & Engineering Decisions

- Layout Engineering (minipage vs. multicols):** Early iterations used the LaTeX multicols environment. However, because the Old Testament contains 39 rows and the New Testament contains 27 rows, multicols automatically attempted to balance column height, causing the Old Testament table to spill over onto Page 6. By refactoring the layout into two independent top-aligned minipage blocks with adjusted line spacing (`\arraystretch{0.93}`), all 66 books fit perfectly onto Page 5.
- Native PDF Action Protocol (/GoToR):** LaTeX's `\href{books/01_Genesis.pdf}{Genesis}` compiles into a native PDF Remote Destination:

$$\text{Action} = \{ '/S': '/GoToR', '/F': 'books/01_Genesis.pdf', '/D': [0, '/Fit'] \}$$

This standard enables desktop PDF viewers (Evince, Okular, Acrobat) to open the relative document at Page 1 with zero external web dependency.

Code Review: Solution C — Cloud Streaming Master Directory

File: `Master_Index_Cloud.tex`

Purpose: Deliver on-the-fly streaming for mobile devices (iPhone/iPad/iCloud) without triggering local sandbox errors or download prompts.

```
1 % Cloud Edition Link Syntax
2 \href{https://cdn.jsdelivr.net/gh/vtrandal/WordofGod@main/books/01_Genesis.pdf}{
  Genesis}
```

The Engineering Challenge: Mobile Sandboxes and Download Traps

When testing `Master_Index.pdf` in iCloud and iOS, two structural obstacles were encountered:

1. **Sandbox Boundary Traversal:** Mobile operating systems isolate documents. Relative links to local subdirectories (`books/01_Genesis.pdf`) are strictly blocked by iOS security policies.
2. **The MIME Type Download Trap:** Direct raw GitHub links (`raw.githubusercontent.com`) intentionally return:

`Content-Type: application/octet-stream`

When mobile Safari or desktop browsers encounter this MIME type, they treat the stream as generic binary data and force a *"Where to save this file?"* dialog.

The Architectural Solution: Multi-CDN Inline Delivery

By directing the LaTeX `\href` endpoints through a high-speed multi-CDN mirror (`cdn.jsdelivr.net`), the HTTP response headers are normalized:

```
1 HTTP/2 200 OK
2 content-type: application/pdf
3 cache-control: public, max-age=604800, s-maxage=43200
4 access-control-allow-origin: *
```

Because `Content-Type: application/pdf` is explicitly served, iOS Safari and desktop viewers render the book ****instantly on the fly**** without download dialogs.

Code Review: Solution D — Digital Web Bookshelf

File: index.html

Purpose: Provide a responsive, zero-dependency visual reading interface with instant search.

```
1 <!-- Real-Time Client-Side Filter Implementation -->
2 <input type="text" id="searchBox" class="search-box"
3     placeholder="Quick search book..." oninput="filterBooks()">
4
5 <script>
6     function filterBooks() {
7         const query = document.getElementById('searchBox').value.trim().toLowerCase();
8         const cards = document.querySelectorAll('.book-card');
9         cards.forEach(card => {
10             const title = card.getAttribute('data-title');
11             card.style.display = title.includes(query) ? 'flex' : 'none';
12         });
13     }
14 </script>
```

Design Highlights

1. **Zero-Dependency Architecture:** Built with pure vanilla HTML5, CSS3, and JavaScript. It does not require Node.js, React, or external CDN frameworks, ensuring 100% offline operation.
2. **Automatic Theme Detection:** Employs CSS media query @media (prefers-color-scheme: dark) with CSS custom properties (--bg, --card-bg, --text-primary) to adapt to user desktop/mobile dark mode preferences automatically.
3. **Dual Action Portal:** The header provides instant action buttons allowing readers to toggle between the Cloud Master Index, the Local Master Index, and the Front Matter.

Code Review: Solution E — Automated Orchestration Engine

File: generate_indexes.py

Purpose: Automate the entire synchronization pipeline across all formats.

```
1 def generate_latex_documents(front_matter, books):
2     # 1. Build Local Offline Edition (Master_Index.tex / .pdf)
3     _build_single_tex(
4         tex_filename="Master_Index.tex",
5         pdf_filename="Master_Index.pdf",
6         link_key="rel_path", # uses books/01_Genesis.pdf
7         ...
8     )
9
10    # 2. Build Cloud Mobile Edition (Master_Index_Cloud.tex / .pdf)
11    _build_single_tex(
12        tex_filename="Master_Index_Cloud.tex",
13        pdf_filename="Master_Index_Cloud.pdf",
14        link_key="cloud_url", # uses CDN application/pdf URLs
15        ...
16    )
```

Design Highlights

1. **Single Source of Truth:** The script iterates through the physical `./books/` directory, inspecting real page counts with `pypdf.PdfReader` rather than relying on hardcoded page tables.
2. **Clean Build Lifecycle:** Spawns `pdflatex` non-interactively (`-interaction=nonstopmode`) and automatically purges transient auxiliary build artifacts (`.aux`, `.log`, `.out`) to maintain a pristine workspace.

Code Review: Solution F — Progressive Web App (PWA) & Audio Engine

Files: manifest.json, sw.js, and index.html

Purpose: Deliver an installable, mobile-optimized scripture reading and audio narration app (reproducing the core “Word of Promise” synchronized audio and text experience) with zero compilation binaries and zero App Store fees.

```

1 // Continuous Chapter Narration Engine with Boundary Verse Tracking
2 function playContinuousAudio(startIdx) {
3   if (synth) synth.cancel();
4   isPlaying = true;
5   document.getElementById('playBtn').innerText = '||';
6
7   // 1. Assemble single continuous string and record character offsets
8   let fullText = "";
9   verseOffsets = [];
10  for (let i = startIdx; i < GENESIS_VERSES.length; i++) {
11    const startChar = fullText.length;
12    fullText += GENESIS_VERSES[i].trim() + " \n\n";
13    verseOffsets.push({ index: i, startChar, endChar: fullText.length });
14  }
15  highlightVerse(startIdx);
16
17  // 2. Play as ONE single continuous utterance (bypasses iOS gesture limits)
18  currentUtterance = new SpeechSynthesisUtterance(fullText);
19  currentUtterance.rate = speechSpeed;
20  currentUtterance.pitch = 0.95;
21  window._activeUtterance = currentUtterance; // GC Anchor
22
23  // 3. Real-time verse tracking via boundary event
24  currentUtterance.onboundary = (e) => {
25    if (!isPlaying) return;
26    for (let r of verseOffsets) {
27      if (e.charIndex >= r.startChar && e.charIndex < r.endChar) {
28        if (currentVerseIndex !== r.index) highlightVerse(r.index);
29        break;
30      }
31    }
32  };
33  currentUtterance.onend = () => stopAudio();
34  synth.speak(currentUtterance);
35 }

```

Design Analysis & Engineering Decisions

- 1. Continuous Chapter Utterance with Boundary Event Tracking:** Solves the critical iOS WebKit limitation where chained programmatic `.speak()` calls inside `onend` callbacks are terminated after two verses due to user-gesture expiration. By assembling the remaining verses into one single continuous string (4,057 characters for Genesis 1, well within the 32k API limit) and mapping character offsets to `onboundary` events, playback runs continuously under the original user tap authorization while maintaining real-time verse auto-scroll centering.
- 2. Standalone Installation without Store Friction:** `manifest.json` specifies the standalone display mode alongside custom high-resolution application icons (`icon-192.png`, `icon-512.png`, and `apple-touch-icon.png`). On iOS Safari, tapping *Add to Home Screen* installs the application

directly to the device springboard, launching in an immersive fullscreen viewport without browser URL bars or navigation buttons.

3. **Dynamic Typography Reflow:** Solves the fixed-canvas “pinch-and-zoom” handicap of mobile PDF viewers. Text dynamically reflows to any viewport, with reader controls for real-time font scaling (A- / A+).
4. **Dual-Tier Service Worker Caching (sw.js):** Implements *Stale-While-Revalidate* for the application shell (HTML/CSS/JS/icons) and *Cache-First* for streaming PDF and audio assets. Once any volume is viewed, the device caches it persistently in `CacheStorage` for guaranteed offline access.
5. **Lock-Screen Media Integration:** Uses the W3C `MediaSession` API to publish book, chapter, and artist metadata to iOS Control Center, CarPlay, and Android lock screens, complete with system play/pause handlers.
6. **iOS In-App Browser Sandbox Traversal:** When the application link is shared through third-party messaging apps (e.g., Facebook Messenger, Slack, Gmail), iOS defaults to opening the target inside a sandboxed `WKWebView`. Apple restricts this in-app environment by suppressing the Safari Share Sheet’s *Add to Home Screen* option. To complete installation, the user must long-press the link in the messaging application and select “*Open in Safari*” (or tap the in-app Safari icon) to transition to native Mobile Safari, restoring the home screen install workflow.
7. **iOS Universal Link Interception Mitigation:** Serving the web application from standard GitHub repository URLs (`github.com/*`) causes iOS to trigger Universal Links, routing the request into the native GitHub developer application. Hosting the PWA on GitHub Pages (<https://vtrandal.github.io/WordofGod/>) provides an isolated web domain that bypasses developer-app hijacking.

Security Architecture & Secrets Governance

Threat Model and Secret Containment

Because this repository connects to a public GitHub remote (`vtrandal/WordofGod`), secret containment was integrated before any commit or push action:

Asset	Storage Mechanism	Governance Rule
API Keys & Tokens	<code>.env</code>	Completely excluded from version control via <code>.gitignore</code> .
Token Credentials	In-Memory / Python Runtime	Pushed via transient auth headers; never hardcoded into Git remote URLs.
Build Artifacts	Local filesystem	Transient LaTeX compilation logs (<code>*.log</code> , <code>*.aux</code>) ignored.

Table 2: Security & Privacy Controls

`.gitignore` Configuration

```

1 # Environment variables & secrets
2 .env
3 *.env
4
5 # LaTeX build artifacts
6 *.aux
7 *.log
8 *.out
9
10 # Temporary troubleshooting media
11 screenshots/

```

Verification Matrix & Quality Assurance

The entire system was subjected to empirical testing across platforms and network conditions:

Test Target	Environment	Result	Observed Behavior
Boundary Slicing	Python 3.12 / Ubuntu	Passed	67 clean PDFs, zero overlapping text.
Local PDF Navigation	Evince / Ubuntu	Passed	/GoToR actions launch local book files.
Raw GitHub HTTP MIME	Safari / iOS	Warning	Returns <code>application/octet-stream</code> (forced download dialog).
CDN Cloud Streaming	Safari / iOS	Passed	Returns <code>application/pdf</code> ; opens on the fly without prompt.
Client-Side Search	Chrome / Firefox	Passed	Sub-10ms DOM response filtering 66 cards.
PWA Home Installation	Mobile Safari / iOS	Passed	Installs to springboard; launches standalone without browser chrome.
In-App Webview Traversal	Messenger / iOS	Mitigated	Webview suppresses install; long-press "Open in Safari" enables install.
Universal Link Routing	GitHub App / iOS	Resolved	GitHub Pages (github.io) bypasses mobile app login intercept.
Synchronized Audio Reader	Safari / iOS & Chrome	Passed	Real-time verse centering with <code>MediaSession</code> lock screen controls.

Table 3: Cross-Platform Quality Assurance Matrix

Conclusion & Engineering Roadmap

Architectural Summary

This engineering initiative demonstrates that monolithic historical documents can be transformed into elegant, modular, cloud-ready software packages with minimal footprint, zero operational costs, and native-grade mobile responsiveness. Through algorithmic boundary extraction, multi-CDN inline delivery, and standards-compliant Progressive Web App technologies, the complete 1769 Authorized King James Bible has been successfully modernized for both offline workstations and cloud-connected mobile devices.

Engineering Roadmap & Future Extensions

Building on the production foundation, the following strategic extensions (formulated in `IDEAS.md`) represent next-phase capabilities:

- Single-File Embedded PDF Package (Idea 1):** Leverage the LaTeX `embedfile` package to pack all 66 standalone book PDFs directly into a single self-contained master document (`Master_Bible_Embedded.pdf`). This provides extreme offline portability without needing a local `./books/` subfolder.
- Single Self-Contained Master Volume with Internal Jumps (Idea 3):** Compile a unified PDF using LaTeX `pdfpages` and internal `\hyperlink` / `\hypertarget` destinations, enabling immediate intra-document jumps between the master directory and biblical chapters with zero filesystem or external protocol dependencies.
- One-Click Offline Library Pre-Caching (Idea 4):** Implement an explicit "Download Entire Bible for Offline Use" caching button within the PWA Service Worker (`sw.js`). Utilizing the

Cache API, the browser will asynchronously pre-fetch and persist all 66 canonical book PDFs into persistent device storage for offline travel.

4. **Extended Multi-Chapter Narration Engine:** Scale the synchronized audio reader beyond Genesis 1 to cover all 1,189 chapters of the Old and New Testaments using public-domain synchronized KJV audio recordings and timed verse timestamps.

END OF DESIGN DOCUMENT